Lecture 6

Thursday, 26 January 2023 10:30 AM

Email purdue username ; Send 2-3 preliminary ideas for term projects

Periodic TISE solution

$V(x+a) = V(x)$

$$\frac{E}{\rho} = \frac{a}{2\pi} \int_{\text{occupied levels}} \frac{\hbar^2}{2m} [k(K)]^2 dK \quad ; \quad K = \frac{2\pi n}{P_a} \quad ; \quad n = 0, \pm 1, \pm 2 \ldots$$

Algorithm to solve:

1) Pick values of K for numerical integration "K points"

2) At each K point, find k(K) from TISE solution or Kohn-Sham (K-S)

3) Advantage: only need TISE / K-S solutions for single unit cell

4) Small lattice constant "a" → larger range of K for integration → need more K points to converge the integral

large "a" → fewer k points

Rule of thumb: # K pts $\gtrsim \frac{2\pi}{0.3a}$ (Each dimension) } K point analysis

Kohn-Sham DFT

Normal DFT: extremely efficient, but kinetic energy functionals aren't good enough (except for low atomic #'s)

∴ Idea: Use a more accurate treatment of KE, at cost of (much) greater computational expense

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Introduced a fictitious, non-interacting gas of e^-'s that has the true, total e^- density $\rho(\underline{r})$ — Slater determinant?

$$\rho(\underline{r}) = \sum_{i=1}^{N} |\psi_i(\underline{r})|^2$$, where $\psi_i(\underline{r})$ are fictitious, single e^- orbitals that don't interact (note: $\rho(\underline{r}) = N \iiint \Psi^* \Psi \, dx_1 \, dx_2 \, dx_3 \ldots dx_N$)

Total Ψ is slater det of ψ_i's (→ anti-symmetric)

KE's for non-interacting particles:

$$\frac{-\hbar^2}{2m} \sum_{i=1}^{N} \int \psi_i^*(\underline{r}) \nabla^2 \psi_i(\underline{r}) \, d\underline{r}$$

{ in PIB: ψ ‖ $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E_{KE}\psi$; $\frac{-\hbar^2}{2m}\int \psi^* \frac{d^2\psi}{dx^2} dx = E_{KE}$ }

Total K-S functional:

$$E[\rho] = \underbrace{-\frac{\hbar^2}{2m} \int \psi_i^*(\underline{r}) \nabla^2 \psi_i(\underline{r}) \, d\underline{r}}_{T_s[\rho]} + \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0} \iint \frac{\rho(\underline{r})\rho(\underline{r}')}{|\underline{r}-\underline{r}'|} d\underline{r} \, d\underline{r}' + \underbrace{E_{xc}[\rho]}_{\text{residual KE excess + exchange + corr.}} + \int V_{ext}(\underline{r}) \rho \, d\underline{r} \quad \ldots\ldots \text{(Eq 1)}$$

Now Lagrangian (w/ constraint of $\int \rho(\underline{r}) \cdot d\underline{r} = N$): $\frac{\partial F[\rho]}{\partial \rho} = \frac{\partial T_s[\rho]}{\partial \rho} + V_{ext}(\underline{r}) + \frac{\partial E_{xc}[\rho]}{\partial \rho} + \frac{e^2}{4\pi\varepsilon_0}\int \frac{\rho(\underline{r}')}{|\underline{r}-\underline{r}'|} d\underline{r}' - \mu$

Defined $V_{eff}(\underline{r}) = V_{ext}(\underline{r}) + \frac{\partial E_{xc}}{\partial \rho} + \frac{e^2}{4\pi\varepsilon_0}\int \frac{\rho}{|r-r'|} \cdot dr'$

$$\frac{\partial T_s[\rho]}{\partial \rho} + V_{eff}(\underline{r}) = \mu$$

$$\frac{\partial T_s[\rho]}{\partial \rho} + V_{ext}(\underline{r}) = \mu$$ Non-interacting systems

Since full TISE separates into single particle equations for non-interacting particles, K+S reasoned this must also be possible for their "fictitious" system

Kohn-Sham eqn's

$$\boxed{\left[-\frac{1}{2}\nabla_i^2 + V_{ext}(\underline{r}) + \frac{\partial E_{xc}}{\partial \rho} + \int \frac{\rho(\underline{r}')}{|\underline{r}-\underline{r}'|} d\underline{r}'\right]\psi_i = \varepsilon_i \psi_i \quad ; \quad i = 1 \cdots N}$$

Algorithm:
1) Guess fictitious ψ_i
2) Find $\rho = \sum |\psi_i|^2$
3) Evaluate $V_{eff}(\underline{r})$ ← Kohn-Sham potential
4) Solve K-S eqs to get new ψ_i's
5) Repeat 1-4

Once converged:
$\Psi(\underline{r}_{all})$ = slater det of ψ_i's

Then find total energy: $E[\rho_{min}] = \sum_{i=1}^{N} \varepsilon_i - \frac{1}{2}\iint \frac{\rho(\underline{r})\rho(\underline{r}')}{|\underline{r}-\underline{r}'|} d\underline{r}\, d\underline{r}' - \int \frac{\partial E_{xc}}{\partial \rho} \rho\, dr + E_{xc}[\rho]$

or use the earlier equation (1)

Example: Two fermions in a 1D box w/ "δ-function repulsion"

K-S energy is: $E[\rho] = -\frac{1}{2}\left(\int_0^L \psi_1^* \frac{d^2\psi}{dx_1^2} dx_1 + \int_0^L \psi_2^* \frac{d^2\rho}{dx_2^2} dx_2\right) + \frac{\alpha}{2}\left(\int_0^L\int_0^L \rho(x)\rho(x')\,\delta(x-x')\, dx_1\, dx_2\right)$

(ignore exchange & correlation) → Can only do this to simple problem → In real case we need to have guesses for these

$\therefore$ K-S Eqs are: $\left[-\frac{1}{2}\frac{d^2}{dx_i^2} + \alpha\rho(x)\right]\psi_i = \varepsilon_i\psi_i$

2 particles: $\left.\begin{aligned} -\frac{1}{2}\frac{d^2\psi_1}{dx_1^2} + \alpha\rho\psi_1 &= \varepsilon_1\psi_1 \\ -\frac{1}{2}\frac{d^2\psi_2}{dx_2^2} + \alpha\rho\psi_2 &= \varepsilon_2\psi_2 \end{aligned}\right\}\ \rho(x) = |\psi_1|^2 + |\psi_2|^2$

Basis expansion: $\psi_1 = \sum_{n=1}^{\infty} a_n \phi_{1n}(x_1) \qquad \psi_2 = \sum_{n=1}^{\infty} b_n \phi_{2n}(x_2)$

PIB solutions: $\phi_n(x) = \sqrt{2}\sin(n\pi x)$ [assume $L=1$]

$$-\frac{1}{2}\frac{d^2}{dx_1^2}\sum_{n=1}^{\infty} a_n \phi_{1n}(x_1) + \alpha\left[\left(\sum_{n=1}^{\infty} a_n\phi_{1n}\right)^2 + \left(\sum b_m \phi_{2m}\right)^2\right]\sum_{j=1}^{\infty} a_j \phi_{1j}(x_1) = \varepsilon_1 \sum_{n=1}^{\infty} a_n \phi_{1n}(x_1)\ ;\ \text{Similar for } x_2$$

Eigenvalue problem (to determine coefficients) & basis functions

$$\Rightarrow -\frac{1}{2}\frac{\pi^2}{2}\sum_{n=1}^{\infty} n^2 a_n \phi_{1n}(x_1) + \quad " \quad "$$

→ Basis functions are orthonormal ∴ Multiply by $\phi_{1k}(x_1)$ and integrate from 0 to 1

$$\frac{\pi^2}{2}k^2 a_k + \alpha\int_0^1 \sum_n\sum_m (a_n a_m + b_n b_m)\,\phi_{1n}\phi_{1m}\underbrace{\sum_{j=1}^{\infty} a_j \phi_{1j}\phi_{1k}}\,dx_1 = \varepsilon_1 a_k$$

represents particle interactions

Similar for ε_2, but don't need to solve that since particles are indistinguishible

Note: $$\int_0^1 \phi_{1n}\phi_{1m}\phi_{1j}\phi_{1k}\,dx = \frac{1}{2}\left[\delta_{n-m-j+k} + \delta_{n-m+j-k} - \delta_{n-m-j-k} - \delta_{n-m+j+k} - \delta_{n+m-j+k} - \delta_{n+m+j-k} + \delta_{n+m-j-k} + \delta_{n+m+j+k}\right]$$

$$\Rightarrow \frac{\pi^2}{2}k^2 a_k + \frac{1}{2}\alpha\sum_n\sum_m (a_n a_m + b_n b_m)\left(a_{n-m+k} + a_{k-n+m} - a_{n-m-k} - a_{m-n-k} - a_{n+m+k} - a_{k-n-m} + a_{n+m-k} + a_{-n-m-k}\right) = \varepsilon_1 a_k\ ;\ \text{Similar for } \varepsilon_2$$

Non-linear eigenvalue solved self-consistently w/ an initial guess for $\{a_i\}$ & $\{b_i\}$

If $n_{max}=2$, $\psi_1^\circ = a_1^\circ \phi_{11}(x_1) + a_2^\circ \phi_{12}(x_1)$

$\psi_2^\circ = b_1^\circ \phi_{21}(x_1) + b_2^\circ \phi_{22}(x_1)$

Plugging ↓ Guess $a_1^\circ = 1$, $a_2^\circ = 0$, $b_1^\circ = 0$, $b_2^\circ = 1$ $\Rightarrow$ $\rho^\circ = a_1^{\circ 2} + b_2^{\circ 2} = 2$

$$\Rightarrow \frac{\pi^2}{2}a_1' + \frac{1}{2}\alpha\left[1\cdot 3a_1' + 1\cdot 2a_1'\right] = \varepsilon_1 a_1' \ ;\ k=1$$

$$2\pi^2 a_2' + \frac{1}{2}\alpha\left[1\cdot 2a_2' + 1\cdot 3a_2'\right] = \varepsilon_1 a_2' \ ;\ k=2$$

from ρ°

$$\frac{\pi^2}{2}a_1 + \frac{5}{2}\alpha a = \varepsilon_1 a_1 \ ;\ 2\pi^2 a_2 + \frac{5}{2}\alpha a_2 = \varepsilon_1 a_2$$

$$\left(\frac{\pi^2}{2} + \frac{5}{2}\alpha - \varepsilon_1\right)a_1 + 0\cdot a_2 = 0 \ ;\ a_1 + \left(2\pi^2 + \frac{5}{2}\alpha - \varepsilon_1\right)a_2 = 0$$

$$\therefore \begin{bmatrix} \frac{\pi^2}{2} + \frac{5}{2}\alpha - \varepsilon_1 & 0 \\ 0 & 2\pi^2 + \frac{5}{2}\alpha - \varepsilon \end{bmatrix} = 0$$

$$\left(\frac{\pi^2}{2} + \frac{5}{2}\alpha - \varepsilon_1\right)\left(2\pi^2 + \frac{5}{2}\alpha - \varepsilon_1\right) = 0$$

$$\therefore \varepsilon_1 = \frac{\pi^2}{2} + \frac{5}{2}\alpha \quad ; \quad \varepsilon_{12} = 2\pi^2 + \frac{5}{2}\alpha$$

Solve for a_1 & a_2:

$$\varepsilon_1 = \frac{\pi^2}{2} + \frac{5}{2}\alpha \rightarrow a_2^1 = 0 \,,\; a_1^1 = 1$$
$$\varepsilon_1 = 2\pi^2 + \frac{5}{2}\alpha \rightarrow a_1^1 = 0 \,,\; a_2^1 = 1$$

} Iteration has converged

$$\therefore \varepsilon = \begin{cases} \frac{\pi^2}{2} + \frac{5}{2}\alpha \\ 2\pi^2 + \frac{5}{2}\alpha \end{cases}$$

$$\Psi_{min}(x_1, x_2) = \frac{1}{\sqrt{2}} \left\{ \phi_{11}(x_1)\,\phi_{21}(x_2) - \phi_{21}(x_2)\,\phi_{12}(x_1) \right\}$$

$$\rho_{min} = 2\sin^2(\pi x) + 2\sin^2(2\pi x)$$

$$E[\rho_{min}] = \frac{5}{2}\pi^2 + \frac{5}{2}\alpha$$

Not the sum one particle

Normal DFT $\Rightarrow \rho_{min} = \frac{N}{L} = 2$ (constant)

$\hbar = m = 1$

$$E[\rho_{min}] = \frac{4}{3}\pi^2 + 2\alpha$$

Exact TFSF: $E_{GS} = \frac{5}{2}\pi^2$

KS more accurate for Al+ atoms

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Al & beyond.

(but DFT approaches KS for smaller atomic sizes)